

CBSE Class 10 Mathematics Standard

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

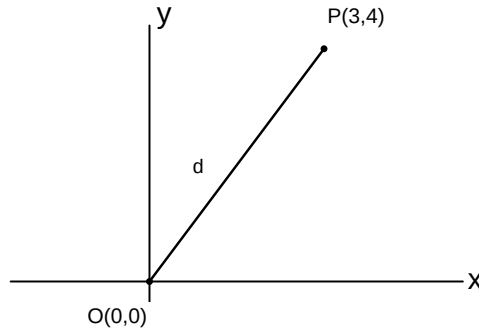
1. This question paper contains 38 questions. All questions are compulsory.
2. Questions are grouped into sections A through E.
3. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. For any positive integer a , Euclid's division lemma states that there exist unique integers q and r such that $a = bq + r$, where r must satisfy which of the following conditions?
(A) (A) $0 < r < b$
(B) (B) $0 < r \leq b$
(C) (C) $0 \leq r < b$
(D) (D) $0 \leq r \leq b$
2. The graph of a polynomial $p(x)$ intersects the x -axis at exactly 3 distinct points. What is the number of zeroes of the polynomial $p(x)$?
(A) (A) 1
(B) (B) 2
(C) (C) 3
(D) (D) 0
3. For the pair of linear equations $x + 2y = 3$ and $2x + 4y = 6$, the lines represented by these equations are:
(A) (A) Intersecting at a unique point
(B) (B) Parallel
(C) (C) Coincident
(D) (D) Intersecting at two points
4. The discriminant of the quadratic equation $2x^2 + 5x - 3 = 0$ is
(A) (A) 1
(B) (B) 19
(C) (C) 49
(D) (D) 31

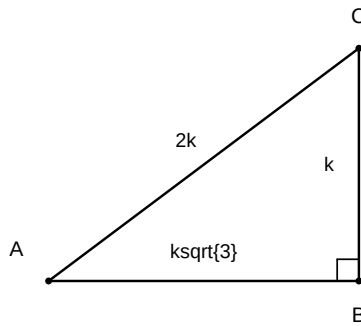
5. If $x = -2$ is one of the roots of the quadratic equation $x^2 + kx - 6 = 0$, then the value of k is
- (A) (A) -1
 - (B) (B) 1
 - (C) (C) 2
 - (D) (D) -2
6. The roots of the quadratic equation $x^2 - 7x + 10 = 0$ are
- (A) (A) 2, -5
 - (B) (B) -2, 5
 - (C) (C) 2, 5
 - (D) (D) -2, -5
7. Which of the following is a factor of the quadratic equation $6x^2 + 11x + 3 = 0$?
- (A) (A) $(3x + 1)$
 - (B) (B) $(3x - 1)$
 - (C) (C) $(2x - 3)$
 - (D) (D) $(6x + 1)$
8. Which of the following constant terms should be added to the expression $x^2 + 8x$ to make it a perfect square trinomial?
- (A) (A) 4
 - (B) (B) 8
 - (C) (C) 16
 - (D) (D) 64
9. In the process of solving $2x^2 - 12x + 5 = 0$ by the method of completing the square, the first step is to divide the entire equation by 2. What is the resulting equation?
- (A) (A) $x^2 - 6x + 2.5 = 0$
 - (B) (B) $x^2 - 12x + 5 = 0$
 - (C) (C) $x^2 - 6x + 5 = 0$
 - (D) (D) $x^2 - 24x + 10 = 0$
10. For a quadratic equation $2x^2 - 5x + k = 0$, if the discriminant is 1, then the value of k is:
- (A) (A) 2
 - (B) (B) 4
 - (C) (C) 3
 - (D) (D) 5

11. The roots of the quadratic equation $x^2 - 4x + 4 = 0$ are:
- (A) (A) 1, 3
 - (B) (B) 2, 2
 - (C) (C) 0, 4
 - (D) (D) 2, -2
12. The roots of the quadratic equation $x^2 - 7x + 12 = 0$ are obtained by factorising the middle term. Which of the following is the correct pair of factors?
- (A) (A) $(x + 3)(x + 4)$
 - (B) (B) $(x - 2)(x - 6)$
 - (C) (C) $(x - 3)(x - 4)$
 - (D) (D) $(x + 3)(x - 4)$
13. If $x = 5$ is one of the roots of the quadratic equation $x^2 - kx - 15 = 0$, then the value of k is:
- (A) (A) 2
 - (B) (B) -2
 - (C) (C) 5
 - (D) (D) -5
14. For an Arithmetic Progression (AP) defined by $a_n = 5n - 3$, the common difference d is:
- (A) (A) 3
 - (B) (B) 5
 - (C) (C) 2
 - (D) (D) 8
15. In $\triangle ABC$, D and E are points on sides AB and AC respectively such that $DE \parallel BC$. If $AD = 4$ cm, $DB = 2$ cm, and $AE = 6$ cm, then the length of EC is:
- (A) (A) 4 cm
 - (B) (B) 2 cm
 - (C) (C) 3 cm
 - (D) (D) 5 cm
16. The distance of the point $P(3, 4)$ from the origin is



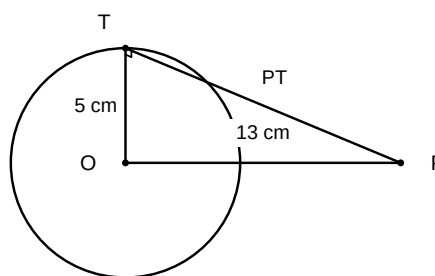
- (A) (A) 3
- (B) (B) 4
- (C) (C) 5
- (D) (D) 7

17. In a right-angled triangle $\triangle ABC$ where $\angle B = 90^\circ$, if $\tan A = \frac{1}{\sqrt{3}}$, then the value of $\sin A$ is:



- (A) (A) $\frac{\sqrt{3}}{2}$
- (B) (B) 1
- (C) (C) $\frac{1}{2}$
- (D) (D) $\frac{1}{\sqrt{2}}$

18. In a circle with centre O , a tangent PT is drawn from an external point P such that the distance $OP = 13 \text{ cm}$. If the radius of the circle is 5 cm , then the length of the tangent PT is:



- (A) (A) 8 cm
- (B) (B) 10 cm
- (C) (C) 12 cm
- (D) (D) 15 cm

19. Assertion (A): The quadratic equation $2x^2 - 5x + 3 = 0$ has two distinct real roots.
Reason (R): A quadratic equation $ax^2 + bx + c = 0$ has two distinct real roots if the discriminant $D = b^2 - 4ac > 0$.

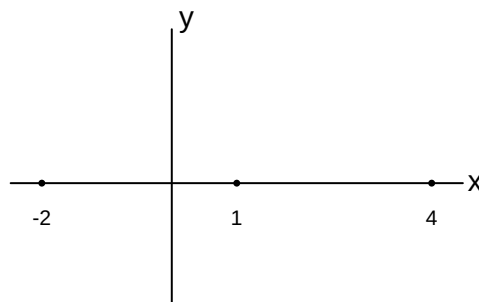
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

20. Assertion (A): The roots of the quadratic equation $x^2 - 7x + 12 = 0$ are 3 and 4.
Reason (R): A quadratic equation $ax^2 + bx + c = 0$ can be solved by splitting the middle term bx into two parts such that their sum is b and their product is ac .

- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

Section B: Very Short Answer (2 marks each)

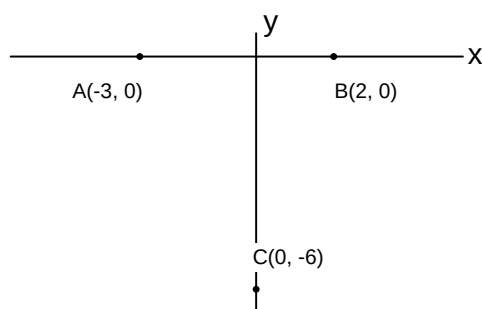
1. Using Euclid's division lemma, show that any positive odd integer is of the form $4q + 1$ or $4q + 3$, where q is some integer.
2. The graph of a polynomial $y = p(x)$ intersects the x -axis at three points: $(-2, 0)$, $(1, 0)$, and $(4, 0)$. State the number of zeros of the polynomial $p(x)$ and write down the values of these zeros.



- Find the discriminant of the quadratic equation $3x^2 - 5x + 2 = 0$ and state the nature of its roots.
- Find the roots of the quadratic equation $x^2 + 5x - 24 = 0$ by the method of factorisation.
- Find the discriminant D of the quadratic equation $3x^2 - 5x + 2 = 0$ and state the nature of its roots.

Section C: Short Answer (3 marks each)

- Use Euclid's division algorithm to find the Highest Common Factor (HCF) of 867 and 255.
- The graph of a polynomial $p(x)$ is given. It is observed that the graph intersects the x -axis at exactly two distinct points $A(-3, 0)$ and $B(2, 0)$ and intersects the y -axis at the point $C(0, -6)$. (i) Based on the geometrical meaning of zeros, state the number of zeros of $p(x)$ and list them. (ii) If the polynomial is of the form $p(x) = ax^2 + bx + c$, find the values of a , b , and c .



- Solve the system of linear equations $x = 2$ and $x - y = -1$ graphically. Also, find the coordinates of the point where these two lines intersect.
- Find the values of k for which the quadratic equation $(k + 1)x^2 - 2(k - 1)x + 1 = 0$ has two real and equal roots.
- Find the roots of the quadratic equation $6x^2 - x - 2 = 0$ by the method of factorisation.
- Solve the quadratic equation $2x^2 - 7x + 3 = 0$ using the quadratic formula.

Section D: Long Answer (4-5 marks each)

- Use Euclid's division lemma to show that the square of any positive integer is either of the form $3m$ or $3m + 1$ for some integer m .

2. The graph of a cubic polynomial $p(x) = ax^3 + bx^2 + cx + d$ intersects the x -axis at exactly three distinct points. The points of intersection are $(-2, 0)$, $(1, 0)$, and $(3, 0)$. Given that the y -intercept of the graph is $(0, 6)$, determine the values of the coefficients a, b, c , and d . Further, find the sum and the product of the zeros of this polynomial.
3. A train travels at a certain average speed for a distance of 540 km. If the speed had been 9 km/h more, it would have taken 1 hour less for the same journey. Form a quadratic equation to represent this situation and find the original average speed of the train.
4. A motorboat whose speed in still water is 18 km/h takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Let x km/h be the speed of the stream. Form a quadratic equation in x and solve it using the factorisation method to find the speed of the stream.

Section E: Case Study (4 marks each)

1. A local NGO is organizing a charity event to distribute food packets. They have two types of delivery routes: Route A, represented by the linear equation $2x + y = 10$, and Route B, represented by the linear equation $x - y = 2$, where x represents the distance covered in km and y represents the fuel consumption in litres. The NGO volunteers need to analyze these routes to optimize their distribution.
 - (i) If we plot the lines $2x + y = 10$ and $x - y = 2$ on a Cartesian plane, what are the coordinates of the point of intersection of these two lines?
 - (ii) At what point does the line $2x + y = 10$ intersect the x -axis?
 - (iii) What is the nature of the lines represented by the pair of equations $x - y = 2$ and $2x - 2y = 4$?
 - (iv) If a new route is introduced as $x + y = k$ and it is known to pass through the point of intersection found in part (i), find the value of k .
 - (i) The point of intersection of $2x + y = 10$ and $x - y = 2$ is:
 - (A) (A) $(4, 2)$
 - (B) (B) $(2, 4)$
 - (C) (C) $(3, 1)$
 - (D) (D) $(5, 0)$
 - (ii) Find the x -intercept of the line $2x + y = 10$.
 - (iii) What is the nature of the lines for the pair $x - y = 2$ and $2x - 2y = 4$?
 - (A) (A) Intersecting
 - (B) (B) Parallel
 - (C) (C) Coincident
 - (D) (D) Perpendicular
 - (iv) If $x + y = k$ passes through $(4, 2)$, what is k ?
2. A group of students is designing a rectangular garden for a school project. The total area of the garden is 300 m^2 . The length of the garden is 7 m more than its width. Let the width of the garden be $x \text{ m}$. Based on this scenario, answer the following questions:

- (i) Which of the following quadratic equations represents the given situation?
- (A) (A) $x^2 + 7x - 300 = 0$
 (B) (B) $x^2 + 7x - 300 = 0$ is incorrect; it is $x^2 + 7x - 300 = 0$ is correct
 (C) (C) $x^2 - 7x - 300 = 0$
 (D) (D) $x^2 + 7x + 300 = 0$
- (ii) Find the discriminant D of the quadratic equation obtained in part (i).
- (iii) Calculate the width of the garden.
- (iv) What is the length of the garden?
- 3.** A local community garden is rectangular in shape. The total area of the garden is 300 m^2 . The length of the garden is 7 m more than its breadth. Let the breadth of the garden be x metres. Based on this information, solve the following questions.
- (i) Which of the following quadratic equations represents the given situation?
- (A) (A) $x^2 + 7x - 300 = 0$
 (B) (B) $x^2 - 7x + 300 = 0$
 (C) (C) $x^2 + 7x - 300 = 0$
 (D) (D) $x^2 - 7x - 300 = 0$
- (ii) Find the roots of the quadratic equation $x^2 + 7x - 300 = 0$ using the factorisation method.
- (iii) What is the breadth of the garden? Justify your answer.
- (iv) Find the length of the garden.